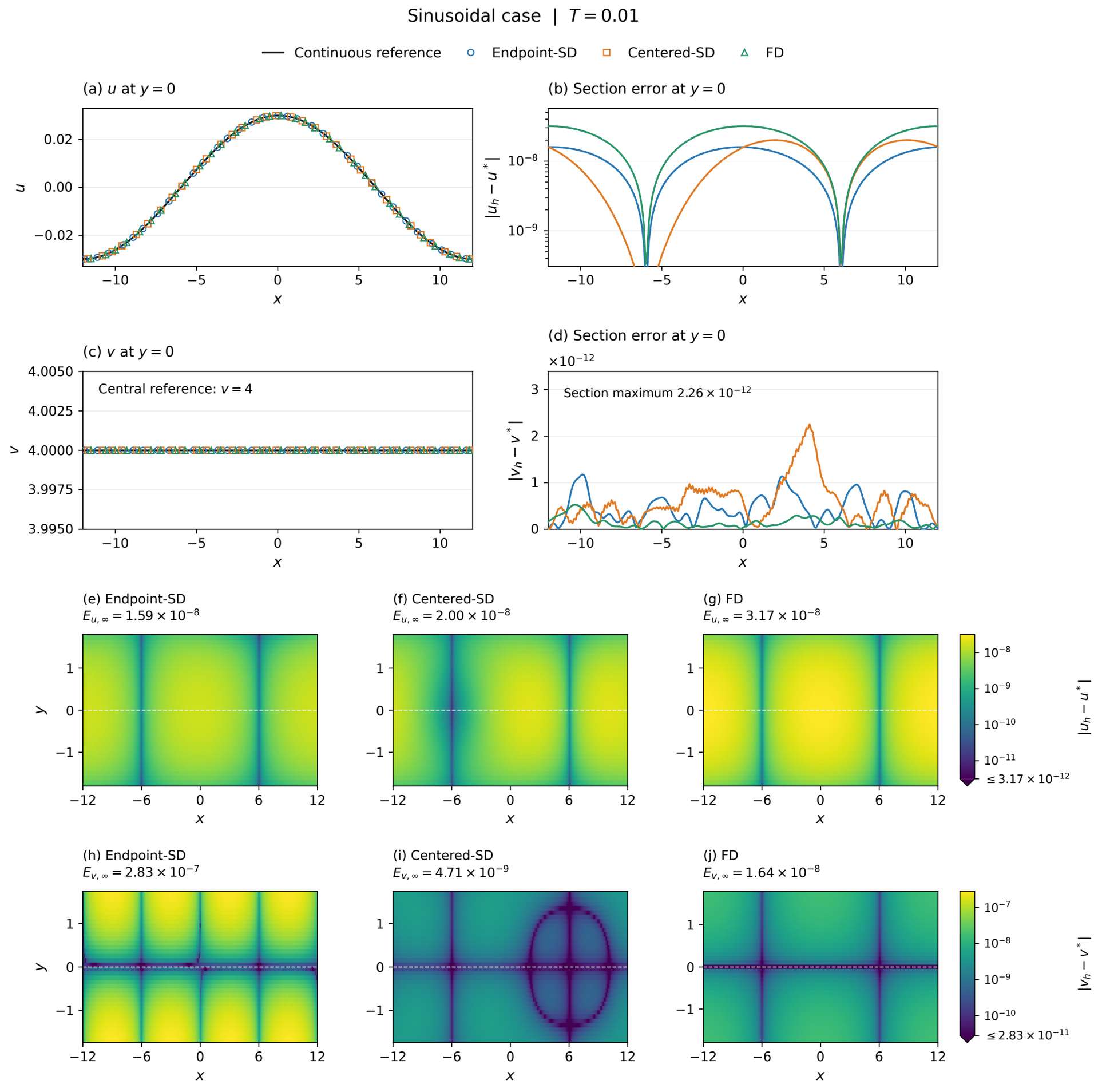

Sinusoidal case | $T = 0.01$
Continuous reference
Endpoint-SD
Centered-SD
FD
(a) u at $y = 0$
u
x
(b) Section error at $y = 0$
$|u_h - u^*|$
(c) v at $y = 0$
Central reference: $v = 4$
v
(d) Section error at $y = 0$
$\times 10^{-12}$
Section maximum 2.26×10^{-12}
$|v_h - v^*|$
(e) Endpoint-SD
$E_{u,\infty} = 1.59 \times 10^{-8}$
(f) Centered-SD
$E_{u,\infty} = 2.00 \times 10^{-8}$
(g) FD
$E_{u,\infty} = 3.17 \times 10^{-8}$
$\leq 3.17 \times 10^{-12}$
(h) Endpoint-SD
$E_{v,\infty} = 2.83 \times 10^{-7}$
(i) Centered-SD
$E_{v,\infty} = 4.71 \times 10^{-9}$
(j) FD
$E_{v,\infty} = 1.64 \times 10^{-8}$
$\leq 2.83 \times 10^{-11}$
y